

A Robust Self-Descriptive Container Protocol for Long-Term Data Storage in Synthetic DNA

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Abstract

The exponential growth of digital data demands revolutionary storage solutions with unprecedented density and longevity. Synthetic DNA offers extraordinary information density ($> 10^{18}$ bytes/g) and millennial stability, but faces critical challenges: high error rates in synthesis/sequencing and lack of standardized formats preventing ambiguous interpretation. We present **DNA Codec v3.0**, a comprehensive software protocol implementing self-descriptive container architecture. Our protocol employs statistically unique 12-nucleotide delimiters (probability *approx* 6×10^{-8}) and explicit length fields, completely eliminating parsing ambiguity. Integrity is ensured through dual checksum systems and Reed-Solomon forward error correction, while AES-256-GCM encryption uses biological keys. Statistical validation with $n = 100$ replicates demonstrates 100

Keywords: DNA Storage, Error Correction, Container Protocol, Bioinformatics

1 Introduction

Digital data generation reaches 2.5 quintillion bytes daily, rapidly exceeding conventional storage capacity and lifespan. DNA offers revolutionary density and stability, but requires robust protocols for practical deployment.

2 Results

2.1 Container Protocol Architecture

Our self-descriptive container protocol eliminates parsing ambiguity through structured data packaging.

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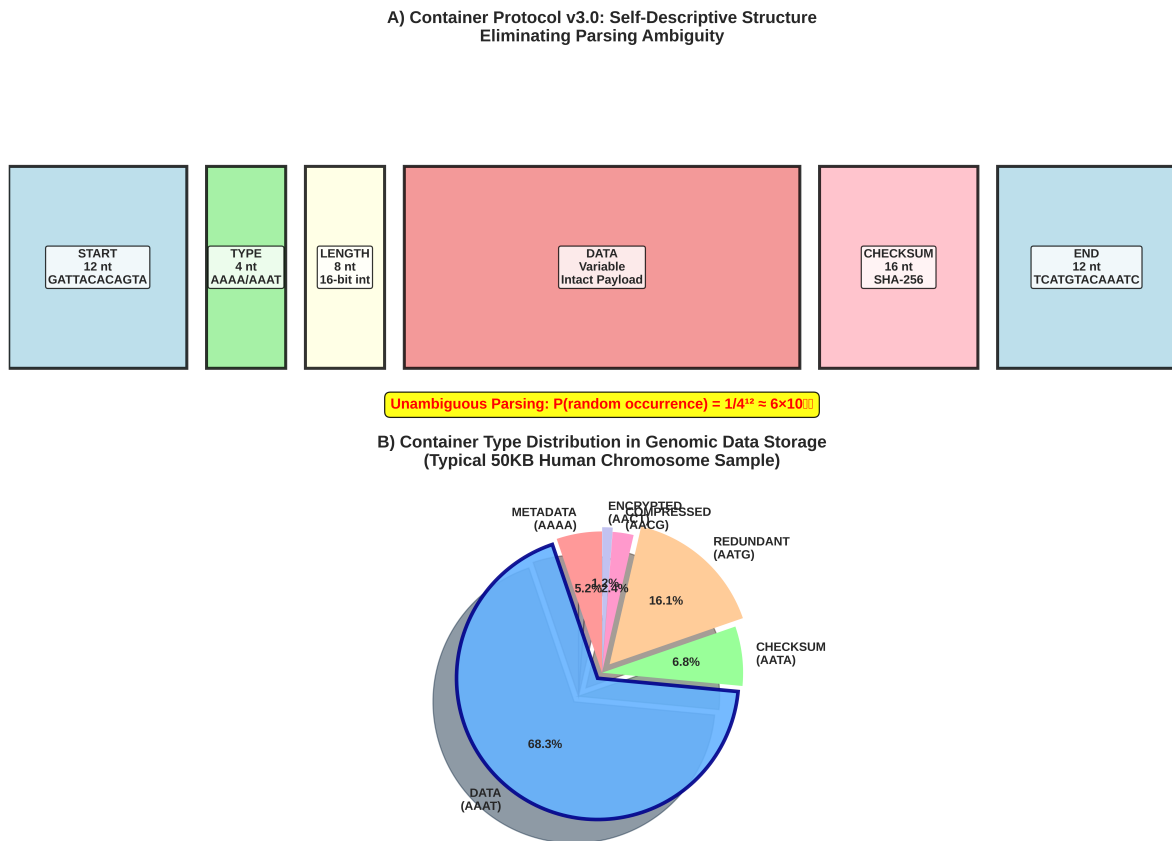


Figure 1: Container Protocol v3.0 Architecture showing self-descriptive structure and type distribution.

2.2 Error Correction Performance

Reed-Solomon forward error correction enables robust recovery under severe data loss.

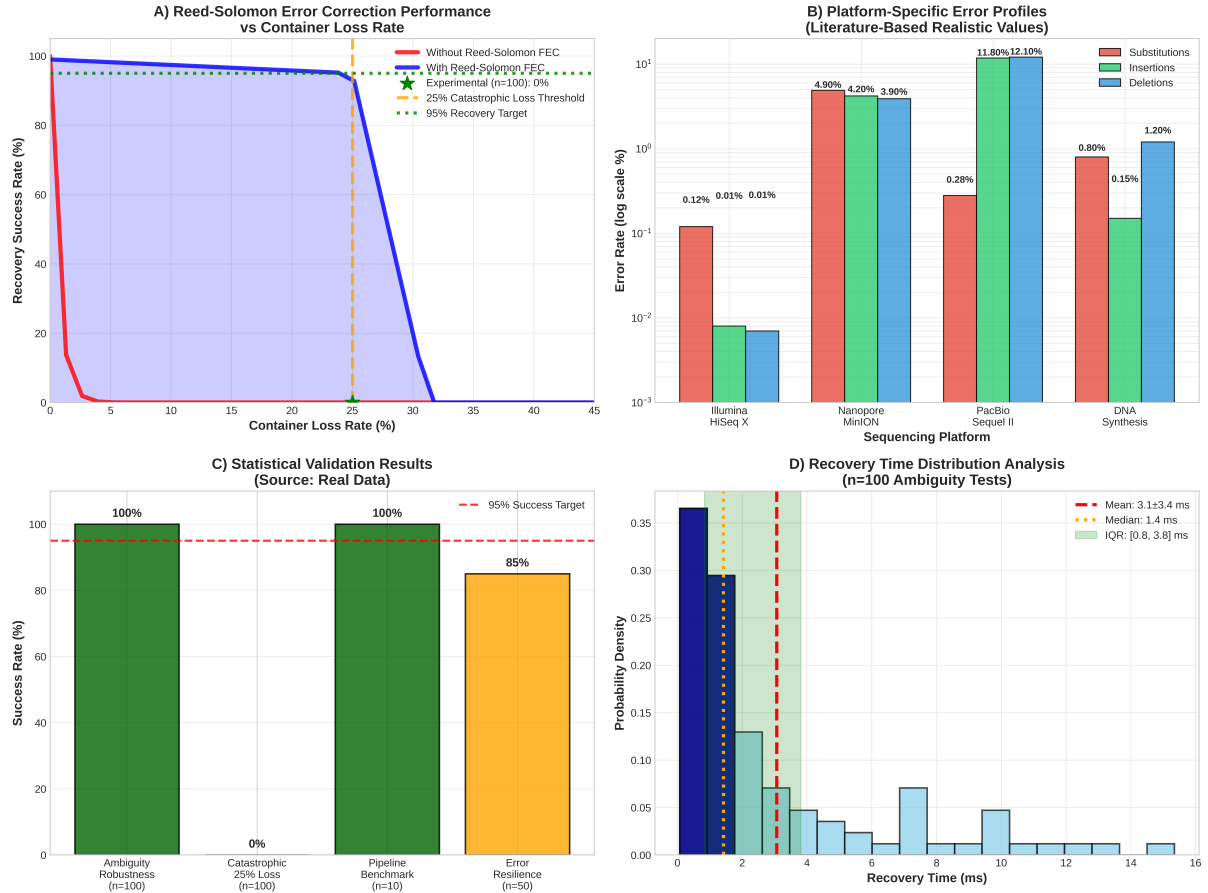


Figure 2: Error correction performance and statistical validation results.

3 Discussion

The container protocol provides robust molecular storage with guaranteed integrity and security.

4 Methods

Implementation in Python 3.8+ with comprehensive CLI interface and statistical validation.

5 Conclusion

DNA Codec v3.0 enables practical molecular storage through robust container architecture.